

EE 596H : Optimization Techniques

Quiz: 90 minutes, 25 marks

Question 1

(10 points)

Consider the problem of minimizing a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2.$$

- (a) Compute the gradient of f at x .
 - (b) Compute the Hessian of f at x .
 - (c) Find points satisfying FONC, if any.
 - (d) Find points satisfying SONC, if any.
 - (e) Find points satisfying SOSC, if any.
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Question 2

(10 points)

Consider the problem

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in \Omega \end{array}$$

where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by $f(x) = 4x_1^2 - x_2^2$, and $\Omega = \{x : x_1^2 + 2x_1 - x_2 \geq 0, x_1 \geq 0, x_2 \geq 0\}$.

- (a) Compute the gradient of f at x .
 - (b) Compute the Hessian of f at x .
 - (c) Does the point $[0, 0]^T$ satisfy the first-order necessary condition?
 - (d) Does the point $[0, 0]^T$ satisfy the second-order necessary condition?
 - (e) Is the point $[0, 0]^T$ a local minimizer of the given problem?
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Question 3

(10 points)

Show that the following functions defined on $\mathbb{R}^n$ are convex:

(a)

$$f(x) = \sum_{k=1}^n e^{x_k}$$

(b)

$$g(x) = \sum_{k=1}^n |x_k|$$

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Question 4

(12 points)

Consider the problem of minimizing $f(x) = 5x_1^2 + 5x_2^2 - x_1x_2 + 11x_1 + 11x_2 + 11$ over $\mathbb{R}^2$.

- (a) Find a global minimizer of f .
 - (b) Starting from the point $[0, 0]^T$, compute two steps of the steepest descent algorithm.
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Question 5

(8 points)

Consider the function $f(x) = x^3 - 6x$ where $x \in \mathbb{R}$.

- (a) Starting at $x = 1$, compute 2 steps of Newton's method.
 - (b) Starting at $x = -1$, compute 2 steps of Newton's method.
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